Project Initialization and Planning Phase

Date	29 June 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

EduGenie is a Google Gemini-powered learning assistant designed to make learning more personalized, interactive, and accessible. It helps students understand concepts, create summaries, practise with quizzes, and receive instant feedback, improving learning confidence and academic performance.

Project Overview	
Objective	The primary objective is to provide learners with personalized, on-demand academic support using Google Gemini to explain concepts, answer questions, and generate practice material.
Scope	The project accepts learner questions and topics, generates AI-assisted explanations, summaries, study plans, and quizzes, and stores learning history and feedback for future reference.
Problem Statement	
Description	Many students struggle to understand difficult subjects because learning materials are generic, doubt clarification is delayed, and individual learning pace is not adequately supported.
Impact	Solving this problem improves concept clarity, increases learner confidence, supports self-paced learning, and helps students prepare more effectively for assignments and examinations.
Proposed Solution	
Approach	Employ Google Gemini API, Python, FastAPI, and Streamlit to deliver personalized learning support through a simple interface, secure API services, and AI-generated educational content.
Key Features	- AI-generated explanations and topic summaries. - Personalized study plans and learning guidance. - Automatic quiz and practice-question generation. - Instant answer evaluation and feedback. - Learning history and saved-topic tracking. - Interactive Streamlit user interface. - REST API documentation through Swagger UI.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 Processor or equivalent
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	FastAPI, Streamlit
Libraries	Additional libraries	Google Generative AI SDK, Pydantic, ReportLab, PyTest, Requests
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	User-provided learning queries and topics; AI-generated explanations and quiz responses; JSON files (history.json, feedback.json)